

A Regression and Visualization Study of Data Science Job Market Trends

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Summary :

The jobs_in_data dataset explores salary trends in the data industry around 10,000 records covering job roles, experience levels, company size, and work settings. It includes key attributes like salary (local and USD), employment type and work environment (remote, hybrid, or in-office).

The goal is to analyze salary differences based on experience, job roles, and company characteristics. A key hypothesis is that senior roles and larger companies offer higher salaries than entry-level jobs or small firms. Using statistical modeling and regression analysis, the dataset helps predict salaries and provides insights for job seekers and employers using the dataset from the years (2020-2023) where we can predict the trends.

Proposed plan:

The analysis of the jobs_in_data dataset will begin with data cleaning and transformation, handling missing values, encoding categorical variables, and standardizing salary values while removing outliers using methods like Z-scores or IQR. Exploratory data analysis (EDA) will include visualizations, such as histograms and box plots, to examine salary distributions across job roles, experience levels, company sizes, and work settings, with correlation analysis revealing key salary factors. A predictive model, like linear regression or decision trees, will be built and evaluated using RMSE and R-squared metrics to assess performance. The findings will provide actionable insights into salary trends, helping job seekers and companies make informed decisions.

Preliminary results:

The dataset has been loaded and pre-processed, with missing values handled through imputation and categorical variables encoded using one-hot encoding. Exploratory data analysis (EDA), including visualizations like histograms, box plots, and density plots, has uncovered notable salary patterns and disparities across various job roles, experience levels, and company sizes. Our initial analysis of the jobs_in_data dataset reveals clear patterns in data science job salaries, with a significant variation in salaries where a small group earns substantially more than the majority. Salaries show trends based on experience levels and company sizes, and when looking at salary trends over time, it appears that salaries tend to increase or remain constant. Additionally, remote jobs are often more competitive in salary, likely due to global hiring practices. Correlation analysis has also identified strong relationships between salary and key factors, confirming the feasibility of the project and setting the stage for further model development.

References:

1. *Data Science Job Salaries* [Dataset]
<https://www.kaggle.com/datasets/ruchi798/data-science-job-salaries>
2. B36050513223. (2022). *Data Science in the Real World: Trends and Challenges* International Journal of Computer Science and Engineering. <https://www.ijscce.org/wp-content/uploads/papers/v13i2/B36050513223.pdf>
3. IEEE. (2022). *Trends in Data Science Career Growth and Salaries* IEEE Xplore. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10859447>
4. Jati. (2022). *Exploring Career Opportunities in Data Science* Jati. https://jati.sites.apiit.edu.my/files/2022/07/Volume6_Issue3_Paper10_2022.pdf
5. SSRN. (2022). *The Impact of Technology Skills on Data Science Salaries* SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3526707